

## 5.9 Cyclic Synchronized Position Mode (CSP)

In this mode, an absolute position preset value is sent to the controller via the fieldbus at fixed intervals (hereinafter referred to as "cycles"). At this point, the controller no longer calculates the ramp, it only follows the preset value. The target position is sent via the PDO and the controller reacts to it immediately. bit 4 in the Controlword does not need to be set (unlike in profile position mode). The target preset value is an absolute value and therefore independent of the number of times each cycle is sent.

### 5.9.1 Structure Plan

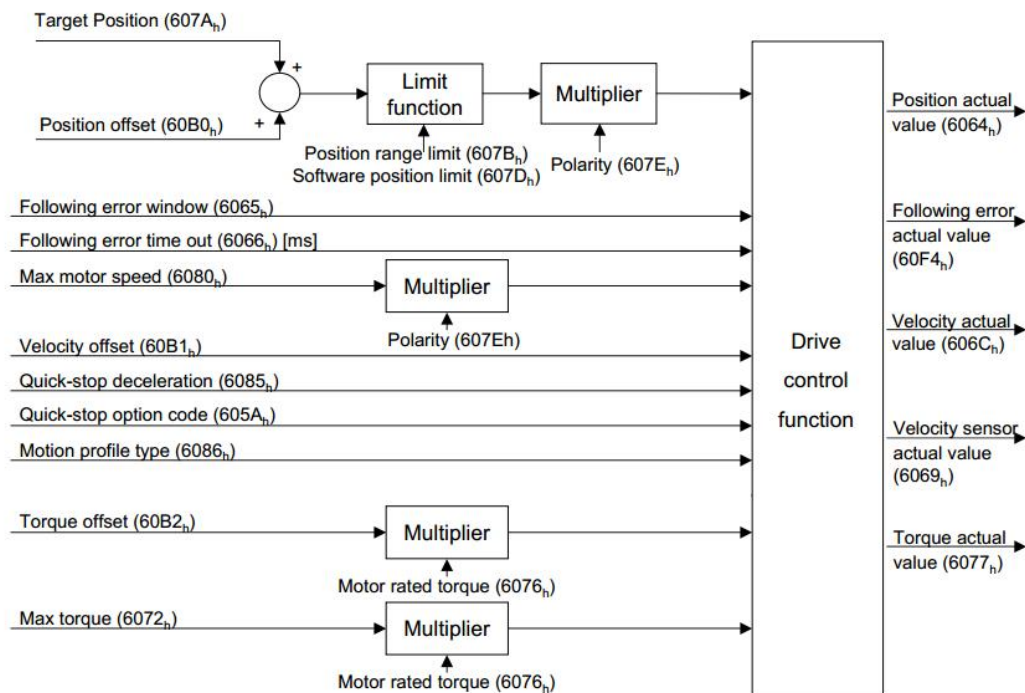


Figure 5- 29

### 5.9.2 Relevant Object

The following objects require attention in this mode:

Table 5- 29

| Object Index                                                                                                       | Description                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 607A <sub>h</sub> (03E7) <sub>h</sub>                                                                              | Preset target position (user units)                                                                                                                                                                                                                                                                     |
| 607D <sub>h</sub> : 01 <sub>h</sub> (03EF) <sub>h</sub><br>607D <sub>h</sub> : 02 <sub>h</sub> (03F1) <sub>h</sub> | Limit values for target position (user units)                                                                                                                                                                                                                                                           |
| 607E <sub>h</sub> (03F3) <sub>h</sub>                                                                              | Direction of rotation (polarity), see "4.3 607E <sub>h</sub> (03F3 <sub>h</sub> ): polarity".                                                                                                                                                                                                           |
| 6065 <sub>h</sub> (03CA) <sub>h</sub>                                                                              | Follow deviation threshold window (user units)                                                                                                                                                                                                                                                          |
| 6066 <sub>h</sub> (03CC) <sub>h</sub>                                                                              | Time range of the following error in milliseconds. If the actual position is out of the error range 6065 <sub>h</sub> (03CA <sub>h</sub> ) for longer than the specified time 6066 <sub>h</sub> (03CC <sub>h</sub> ), the follow error will be triggered and bit13 of the control word will be set to 1 |
| 6080 <sub>h</sub> (03F6) <sub>h</sub>                                                                              | Maximum speed of motor (rpm)                                                                                                                                                                                                                                                                            |
| 6085 <sub>h</sub> (0400) <sub>h</sub>                                                                              | Stopping deceleration rate when performing a "quick stop" (user                                                                                                                                                                                                                                         |

|                                       |                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | unit/s <sup>2</sup> )                                                                                                                                                                                                                                                                                                                                  |
| 605A <sub>h</sub> (03BF) <sub>h</sub> | Quick stop mode selection                                                                                                                                                                                                                                                                                                                              |
| 605D <sub>h</sub> (03C0) <sub>h</sub> | Pause mode selection                                                                                                                                                                                                                                                                                                                                   |
| 6086 <sub>h</sub> (0402) <sub>h</sub> | The type of ramp to be traveled, if the value is "0", the shock will not be limited, if the value is "3", the shock will be limited with the value from 60A4 <sub>h</sub> : 01 <sub>h</sub> - 04 <sub>h</sub> (041E <sub>h</sub> ~0424 <sub>h</sub> ), this device only uses 01 <sub>h</sub> -02 <sub>h</sub> (041E <sub>h</sub> ~0420 <sub>h</sub> ). |
| 60B0 <sub>h</sub> (0426) <sub>h</sub> | Position offset (user units)                                                                                                                                                                                                                                                                                                                           |
| 60B1 <sub>h</sub> (0428) <sub>h</sub> | Speed bias (user units)                                                                                                                                                                                                                                                                                                                                |
| 6064 <sub>h</sub> (03C8) <sub>h</sub> | Feedback on the current absolute user position of the motor (user units)                                                                                                                                                                                                                                                                               |
| 606C <sub>h</sub> (03D5) <sub>h</sub> | Current actual speed feedback value (rpm)                                                                                                                                                                                                                                                                                                              |
| 60F4 <sub>h</sub> (0440) <sub>h</sub> | Real-time following deviation (user units)                                                                                                                                                                                                                                                                                                             |

### 5.9.3 Control Commands and Status Messages

#### start

To enable this mode, the value "0" must be set in object 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) and "8" in object 6060<sub>h</sub> (03C2<sub>h</sub>).

#### control word

The bits in object 6040<sub>h</sub> (0380<sub>h</sub>) have no special function in cyclic synchronized position mode, see section 3.1.2 Control words for the definition of the general-purpose bits.

**Note: CSP mode only supports absolute position commands**

#### status word

|             |                 |                         |             |          |             |   |     |
|-------------|-----------------|-------------------------|-------------|----------|-------------|---|-----|
| 15          | 14              | 13                      | 12          | 11       | 10          | 9 | 0   |
| (see 3.1.2) | Following error | Target position ignored | (see 3.1.2) | reserved | (see 3.1.2) |   |     |
| MSB         |                 |                         |             |          |             |   | LSB |

In cyclic synchronized position mode, the following bits in object **6041<sub>h</sub>** (0381<sub>h</sub>) have a special function:

Table 5- 30

| Bit | Value | Description                                                                                                                   |
|-----|-------|-------------------------------------------------------------------------------------------------------------------------------|
| 12  | 0     | The controller does not follow the target preset, and the preset value of 607A <sub>h</sub> (03E7 <sub>h</sub> ) is ignored.  |
|     | 1     | The controller follows the target preset and object 607A <sub>h</sub> (03E7 <sub>h</sub> ) is used as position control input. |
| 13  | 0     | No following error                                                                                                            |
|     | 1     | With following error                                                                                                          |

### 5.9.4 Functional Description

In this mode, an absolute position preset value is sent to the controller via the fieldbus at fixed intervals (hereinafter referred to as "cycles").

### 5.9.5 Configuration Example

- Configuration mode:  
Working in cyclic synchronized position mode: write 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) = 0, run mode 6060<sub>h</sub> (03C2<sub>h</sub>) = 0x08;  
Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)  
Send : 01 10 03 C2 00 01 02 00 08 (set to cyclic synchronized position mode)
- The motor runs:  
Write control word 6040<sub>h</sub> (0380<sub>h</sub>) = 0x06→0x07→0x0F;  
Send: 01 10 03 80 00 01 02 00 06 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x6 to make motor ready)  
Send: 01 10 03 80 00 01 02 00 07 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x7 to disable motor)  
Send: 01 10 03 80 00 01 02 00 0F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0xF to enable motor)
- The host computer sends the target position according to the synchronization cycle 607A<sub>h</sub> (03E7<sub>h</sub>)

## 5.10 Cyclic Synchronized Velocity Mode (CSV)

In the cyclic synchronized speed mode, the upper controller sends the calculated target speed 60FF<sub>h</sub> (0448<sub>h</sub>) to the driver in a cyclic synchronized manner, and the speed and torque adjustments are executed internally.

### 5.10.1 Structure Plan

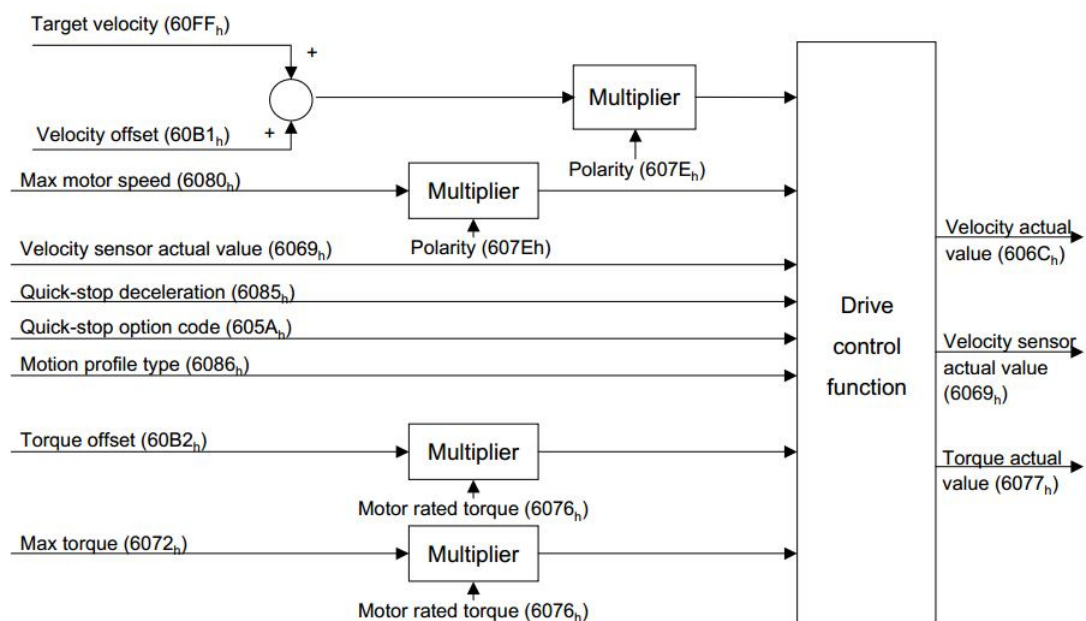


Figure 5- 30

### 5.10.2 Relevant Object

The following objects require attention in this mode:

Table 5- 31

| Object Index                          | Description                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 605A <sub>h</sub> (03BF) <sub>h</sub> | Quick stop mode selection                                                                                                                                                                                                                                                                                                                                                                                             |
| 605D <sub>h</sub> (03BE) <sub>h</sub> | Pause mode selection                                                                                                                                                                                                                                                                                                                                                                                                  |
| 606C <sub>h</sub> (03D5) <sub>h</sub> | Current actual speed feedback value (in rpm)                                                                                                                                                                                                                                                                                                                                                                          |
| 607E <sub>h</sub> (03F3) <sub>h</sub> | Direction of rotation (polarity), see "4.3 607E <sub>h</sub> : Polarity".                                                                                                                                                                                                                                                                                                                                             |
| 60FF <sub>h</sub> (0448) <sub>h</sub> | Setting the target speed (user unit/s)                                                                                                                                                                                                                                                                                                                                                                                |
| 6080 <sub>h</sub> (03F6) <sub>h</sub> | Maximum speed of motor (rpm)                                                                                                                                                                                                                                                                                                                                                                                          |
| 6085 <sub>h</sub> (0400) <sub>h</sub> | Stopping deceleration rate when performing a "quick stop" (user unit/s <sup>2</sup> )                                                                                                                                                                                                                                                                                                                                 |
| 6086 <sub>h</sub> (0402) <sub>h</sub> | <p>Select the type of curve, i.e. slope.</p> <p>If the value is "0", the shock (plus acceleration) is not limited, i.e., the trapezoidal curve;</p> <p>If the value is "3", the impact will be limited by the value in 60A4<sub>h</sub> : 01<sub>h</sub> - 04<sub>h</sub> (041E<sub>h</sub> ~0424<sub>h</sub>), only 01<sub>h</sub> -02<sub>h</sub> (041E<sub>h</sub> ~0420<sub>h</sub>) is used for this device.</p> |
| 60B1 <sub>h</sub> (0428) <sub>h</sub> | Position offset (user units)                                                                                                                                                                                                                                                                                                                                                                                          |
| 60B2 <sub>h</sub> (042A) <sub>h</sub> | Torque bias (0.1%)                                                                                                                                                                                                                                                                                                                                                                                                    |
| 6072 <sub>h</sub> (03DC) <sub>h</sub> | Maximum torque (0.1%) on the entire ramp (acceleration, torque conservation, braking)                                                                                                                                                                                                                                                                                                                                 |

### 5.10.3 Control Commands and Status Messages

#### start

To enable this mode, the value "0" must be set in object 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) and "9" in object 6060<sub>h</sub> (03C2<sub>h</sub>).

#### control word

In cyclic synchronized speed mode, the bits in the object 6040<sub>h</sub> (0380<sub>h</sub>) have no special function, see section 3.1.2 Control words for the definition of the general-purpose bits.

#### status word

| 15 14       | 13       | 12                      | 11          | 10       | 9 0         |
|-------------|----------|-------------------------|-------------|----------|-------------|
| (see 3.1.3) | reserved | Target velocity ignored | (see 3.1.3) | reserved | (see 3.1.3) |
| MSB         |          |                         |             |          | LSB         |

In cyclic synchronized speed mode, the following bits in object 6041<sub>h</sub> (0381<sub>h</sub>) have a special function:

Table 5- 32

| Bit | Value | Definition                    |
|-----|-------|-------------------------------|
| 12  | 0     | Target speed will be rounded  |
|     | 1     | Target speed as planner input |

### 5.10.4 Functional Description

In the cyclically synchronized speed mode, the upper controller sends the calculated target speed 60FF<sub>h</sub> (0448<sub>h</sub>) to the motor driver in a cyclically synchronized manner, and the speed regulation is performed internally by the motor.

### 5.10.5 Configuration Example

- Setup Mode:

Write 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) = 0, Run Mode 6060<sub>h</sub> (03C2<sub>h</sub>) = 0x09 to operate in Cyclic Synchronized Speed Mode;

Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)

Send: 01 10 03 C2 00 01 02 00 09 (set to cyclic synchronized speed mode)

- Write control words

6040<sub>h</sub> (0380<sub>h</sub>) = 0x06→0x07→0x0F;

Send: 01 10 03 80 00 01 02 00 06 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x6 to make motor ready)

Send: 01 10 03 80 00 01 02 00 07 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x7 to disable motor)

Send: 01 10 03 80 00 01 02 00 0F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0xF to enable motor)

The motor is running and the host computer needs to write the speed object according to the synchronization cycle 60FF<sub>h</sub> (0448<sub>h</sub>)

## 5.11 Cyclic Synchronous Torque Mode (CST)

In this mode, the upper controller sends the calculated target torque 6071<sub>h</sub> (03DB<sub>h</sub>) to the drive periodically and synchronously, and the torque regulation is performed internally. When the speed reaches the limit value it will enter the speed regulation phase.

### 5.11.1 Structure Plan

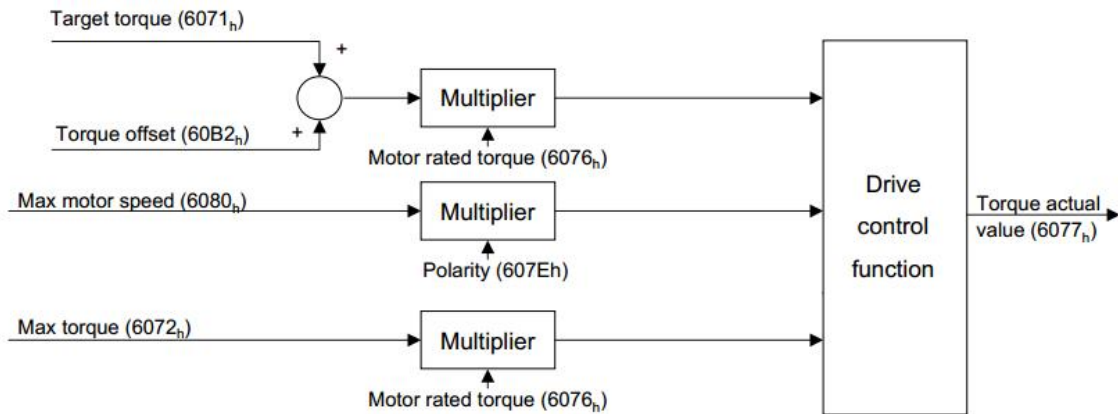


Figure 5- 31

### 5.11.2 Relevant Object

The following objects require attention in this mode:

Table 5- 33

| Object Index                          | Description                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------------|
| 605A <sub>h</sub> (03BF) <sub>h</sub> | Quick stop mode selection                                                             |
| 605D <sub>h</sub> (03BE) <sub>h</sub> | Pause mode selection                                                                  |
| 6071 <sub>h</sub> (03DB) <sub>h</sub> | Setting target torque (0.1%)                                                          |
| 6072 <sub>h</sub> (03DC) <sub>h</sub> | Maximum torque (0.1%) on the entire ramp (acceleration, torque conservation, braking) |
| 6073 <sub>h</sub> (03DD) <sub>h</sub> | Maximum allowable current (0.01A)                                                     |
| 6080 <sub>h</sub> (03F6) <sub>h</sub> | Maximum speed of motor (rpm)                                                          |
| 60B2 <sub>h</sub> (042A) <sub>h</sub> | Torque bias (0.1%)                                                                    |
| 6088 <sub>h</sub> (0405) <sub>h</sub> | Torque ramp type (0-slope, 2-none)                                                    |
| 6077 <sub>h</sub> (03E3) <sub>h</sub> | Torque actual value (0.1%)                                                            |

### 5.11.3 Control Commands and Status Messages

#### start

To enable this mode, the value "0" must be set in object 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) and "10" in object 6060<sub>h</sub> (03C2<sub>h</sub>).

#### control word

In cyclic synchronous torque mode, the bits in object 6040<sub>h</sub> (0380<sub>h</sub>) have no special function; see section 3.1.2 Control words for the definition of the general-purpose bits.

### status word

|             |          |                       |             |          |             |   |   |
|-------------|----------|-----------------------|-------------|----------|-------------|---|---|
| 15          | 14       | 13                    | 12          | 11       | 10          | 9 | 0 |
| (see 3.1.3) | reserved | Target torque ignored | (see 3.1.3) | reserved | (see 3.1.3) |   |   |
| MSB         |          |                       |             | LSB      |             |   |   |

In cyclic synchronous torque mode, the following bits in object **6041<sub>h</sub>** (0381<sub>h</sub>) have a special function:

Table 5- 34

| Bit | Value | Definition                      |
|-----|-------|---------------------------------|
| 12  | 0     | Target torque will be discarded |
|     | 1     | Target torque as planner input  |

#### 5.11.4 Functional Description

In this mode, the upper controller sends the calculated target torque 6071<sub>h</sub> (03DB<sub>h</sub>) to the driver in a cyclically synchronized manner.

#### 5.11.5 Configuration Example

- Setup Mode:  
Write 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) = 0, run mode 6060<sub>h</sub> (03C2<sub>h</sub>) = 0x0A to operate in cyclic synchronous torque mode;  
Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)  
Send : 01 10 03 C2 00 01 02 00 0A (set to cyclic synchronous torque mode)
- Write control words  
6040<sub>h</sub> (0380<sub>h</sub>) = 0x06→0x07→0x0F, motor running, upper computer needs to write target torque 6071<sub>h</sub> (03DB<sub>h</sub>)  
Send: 01 10 03 80 00 01 02 00 06 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x6 to make motor ready)  
Send: 01 10 03 80 00 01 02 00 07 (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0x7 to disable motor)  
Send: 01 10 03 80 00 01 02 00 0F (set 6040<sub>h</sub> (0380<sub>h</sub>) to 0xF to enable motor)

## 5.12 NiMotion mode

### 5.12.1 Summary

The NiMotion mode is different from the CIA402 mode in that there is no state machine management, and the position, speed and torque are controlled through the parameters configured in the multi-function terminals, including the NiMotion position mode, the NiMotion speed mode and the NiMotion torque mode. 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) (control mode selection) is used to determine which operating mode is in operation. See 5.1 Overview of Control Modes for specific correspondences.

## 5.13 NiMotion Position Mode

### 5.13.1 Structure Plan

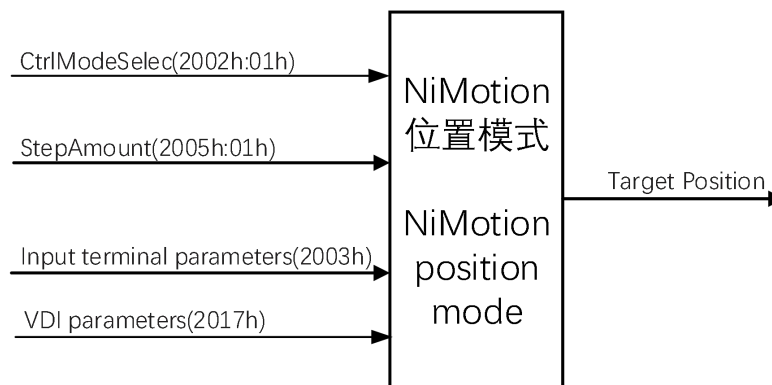


Figure 5- 32

### 5.13.2 Relevant Object

The following objects require attention in this mode:

Table 5- 35

| Object Index                                              | Description                                                                                                                                                                                                                        |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2002 <sub>h</sub> :01 <sub>h</sub> (00B1 ) <sub>h</sub>   | Operation mode selection                                                                                                                                                                                                           |
| 2005 <sub>h</sub> :01 <sub>h</sub> (010D ) <sub>h</sub>   | Position command source, 0-pulse, 1-step, 2-multi-segment position                                                                                                                                                                 |
| 2005 <sub>h</sub> :05 <sub>h</sub> (0112 ) <sub>h</sub>   | Step amount setting (encoder unit)                                                                                                                                                                                                 |
| 2017 <sub>h</sub> (0326 <sub>h</sub> ~0346 ) <sub>h</sub> | Functional configuration of the virtual input terminals (see: "6.1.1 Digital input")                                                                                                                                               |
| 2003 <sub>h</sub> (00D5 <sub>h</sub> ~00E6 ) <sub>h</sub> | Functional configuration of the physical input terminals (see: "6.1.1 Digital input")<br>Note: You cannot have multiple input terminals configured as the same switch at the same time, or you will report a terminal setup fault. |

### 5.13.3 Control Commands and Status Messages

There is no state machine management in this mode, the motor is enabled via the terminals.



## Enable

Prerequisite for this mode of operation: the integrated low-voltage servo motor is switched to NiMotion position mode (2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) write 1).

### 1 Location Source Selection:

The object (2005<sub>h</sub> : 01<sub>h</sub> (010D<sub>h</sub>)) is a source of position instructions, meaning: 0-pulse, 1-step amount;

### 2 Setting the multi-function terminals (both physical and virtual terminals are applicable, see "6.1 Digital Inputs and Outputs" for details on terminal settings).

If the set position instruction source is 1, configure one input terminal as motor enable and configure its valid logic level according to the actual situation, the default is low and valid; set the other input terminal as step amount enable and configure its valid logic level according to the actual situation, the default is low and valid;

If the source of the set position command is 0 (external pulse given), in addition to setting the terminals for motor enable, it is also necessary to set the input terminals DI1 and DI2 for the corresponding pulse input function.

### 3 Position Setting

If the source of the position instruction is selected as 1, you need to set the step amount given value to be written to object 2005<sub>h</sub> : 05<sub>h</sub> (0112<sub>h</sub>);

If the source of the position command is selected as 0, a pulse signal from input terminals DI1 and DI2 is required.

### 4 Run Run

If you select the position command source as 1, enable the motor through the terminal, and then enable the step amount signal through the terminal, the motor travels the distance (unit: encoder unit) set by 2005<sub>h</sub> : 05<sub>h</sub> (0112<sub>h</sub>), and the motor travels once for each enabled step amount;

If the source of position command is selected as 0, motor enable is performed according to the terminal and the motor runs according to the pulse signal from the input terminal.

## 5.13.4 Functional Description

In this mode, the motor will travel a controlled distance based on the input signal from the terminal.

## 5.13.5 Example

### Stepping volume given position control

Parameter Settings:

- Set to NiMotion position mode 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>)=1  
Send: 01 10 00 B1 00 01 02 00 01 (set to NiMotion position mode)
- Setting the position instruction source as a step amount: 2005<sub>h</sub> : 01<sub>h</sub> (010D<sub>h</sub>)=1

Send: 01 10 01 0D 00 01 02 00 01 (Setting the source of the position command as a step)

- Set the step amount: 2005<sub>h</sub> : 05<sub>h</sub> (0112<sub>h</sub>) = 50

Send: 01 10 01 12 00 01 02 00 32 (set step amount to 50)

- Set motor enable: 2003<sub>h</sub> : 03<sub>h</sub> (00D5<sub>h</sub>) = 1, logic active low: 2003<sub>h</sub> : 04<sub>h</sub> (00D6<sub>h</sub>) = 0

Send: 01 10 00 D5 00 01 02 00 01 (sets DI1 to step enable)

Send: 01 10 00 D6 00 01 02 00 00 (sets DI1 logic active low)

- Setting the step amount enable: 2003<sub>h</sub> : 05<sub>h</sub> (00D7<sub>h</sub>) = 20, Logic is valid for falling edge: 2003<sub>h</sub> : 06<sub>h</sub> (00D8<sub>h</sub>) = 3

Send: 01 10 00 D7 00 01 02 00 14 (sets DI2 to step enable)

Send : 01 10 00 D8 00 01 02 00 03 (sets DI2 logic to be active on falling edge)

Running:

- Motor enable: DI1 input low level
- Step amount enable: DI2 input a falling edge, step a distance; each input a falling edge will step a distance

### Orthogonal pulse-given position control

Parameter Settings:

- Set to NiMotion position mode 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) = 1

Send: 01 10 00 B1 00 01 02 00 00 (set to NiMotion position mode)

- Setting the position command source to pulse: 2005<sub>h</sub> : 01<sub>h</sub> (010D<sub>h</sub>) = 0

Send: 01 10 01 0D 00 01 02 00 00 (set position command source to pulse)

- Set the input terminals:

DI1 Quadrature pulse input A: 2003<sub>h</sub> : 03<sub>h</sub> (00D5<sub>h</sub>) = 40

Send: 01 10 00 D5 00 01 02 00 28 (sets DI1 to quadrature input pulse A)

DI2 Quadrature Pulse Input B: 2003<sub>h</sub> : 05<sub>h</sub> (00D7<sub>h</sub>) = 41

Send: 01 10 00 D7 00 01 02 00 29 (sets DI2 to quadrature input pulse B)

DI3 function number is set to motor enable:

Function selection for step enable: 2003<sub>h</sub> : 07<sub>h</sub> (00D9<sub>h</sub>) = 1

Send: 01 10 00 D9 00 01 02 00 01 (sets DI3 to step enable)

Logic selection active low: 2003<sub>h</sub> : 08<sub>h</sub> (00DA<sub>h</sub>) = 0

Send: 01 10 00 DA 00 01 02 00 00 (sets DI3 logic active low)

Running:

- Motor enable: DI3 input low level
- Pulse input control: Terminals DI1 and DI2 input orthogonal pulses and the motor is controlled in position according to the input pulses.

## 5.14 NiMotion Speed Mode

### 5.14.1 Structure Plan

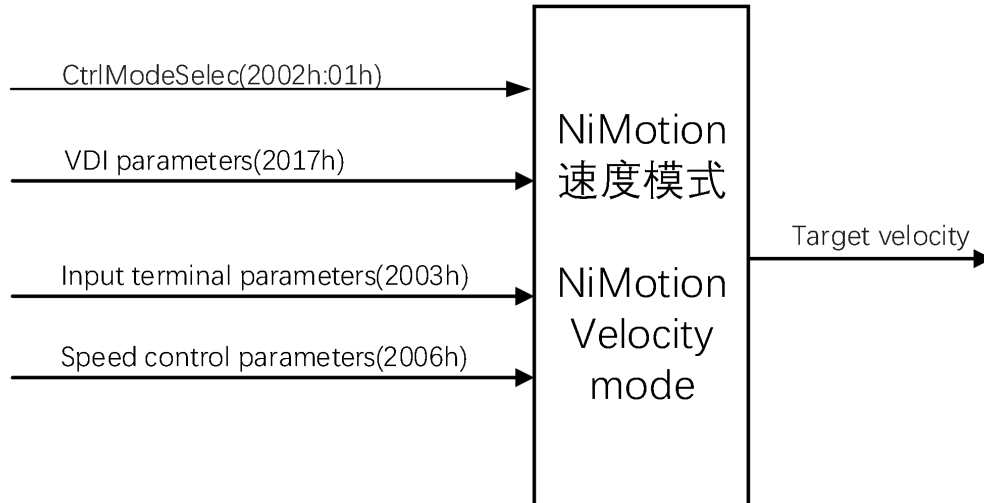


Figure 5- 33

### 5.14.2 Relevant Object

The following objects require attention in this mode:

Table 5- 36

| Object Index                                                           | Description                                                                                                                                                                                                                        |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2000 <sub>h</sub> : 0F <sub>h</sub> (006B <sub>h</sub> ) <sub>h</sub>  | Maximum speed, rpm                                                                                                                                                                                                                 |
| 2002 <sub>h</sub> : 01 <sub>h</sub> (00B1 <sub>h</sub> ) <sub>h</sub>  | Operation mode selection                                                                                                                                                                                                           |
| 2006 <sub>h</sub> : 01 <sub>h</sub> (0148 <sub>h</sub> )               | NiMotion speed mode master speed feed source, 0-digital feed, 3-duty cycle feed (20Hz-20kHz, 1kHz recommended)                                                                                                                     |
| 2006 <sub>h</sub> : 02 <sub>h</sub> (0149 <sub>h</sub> )               | NiMotion speed mode auxiliary speed feed source, 0 - digital feed, 3 - multispeed                                                                                                                                                  |
| 2006 <sub>h</sub> : 03 <sub>h</sub> (014A <sub>h</sub> )               | NiMotion speed mode speed command selection, 0-primary speed given, 1-secondary speed given, 2-primary and secondary speed given (default 0-primary speed given)                                                                   |
| 2006 <sub>h</sub> : 04 <sub>h</sub> (014B <sub>h</sub> )               | Speed command numerical given value in rpm                                                                                                                                                                                         |
| 2006 <sub>h</sub> : 07 <sub>h</sub> (014E <sub>h</sub> )               | The speed command acceleration ramp time constant, in ms, indicates the time taken to accelerate from 0 rpm to 1000 rpm.                                                                                                           |
| 2006 <sub>h</sub> : 08 <sub>h</sub> (014F <sub>h</sub> )               | Speed command deceleration ramp time constant in ms                                                                                                                                                                                |
| 2017 <sub>h</sub> (0326 <sub>h</sub> ~0346 <sub>h</sub> ) <sub>h</sub> | Functional configuration of the virtual input terminals (see: "6.1.1 Digital input")                                                                                                                                               |
| 2003 <sub>h</sub> (00D5 <sub>h</sub> ~00E6 <sub>h</sub> ) <sub>h</sub> | Functional configuration of the physical input terminals (see: "6.1.1 Digital input")<br>Note: You cannot have multiple input terminals configured as the same switch at the same time, or you will report a terminal setup fault. |

### 5.14.3 Control Commands and Status Messages

There is no state machine management in this mode, the motor is enabled via the terminals.

#### Enable

Prerequisite for this mode of operation: the integrated low-voltage servo motor is switched to NiMotion speed mode (2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) set to 2).

#### 1 Speed source selection:

The object is the speed instruction source 2006<sub>h</sub> : 01<sub>h</sub> (0148<sub>h</sub>), meaning: 0 - number given, 3 - duty cycle given;

#### 2 Set the multi-function terminals (both physical and virtual terminals are applicable, see "6.1 Digital Inputs and Outputs" for details on terminal settings):

Configure one input terminal as motor enable, and configure its valid logic level according to the actual situation, the default low level is valid. If the source of speed command set in the previous step is 3 (duty cycle given), you need to set the corresponding input terminals DI1 and DI2 for duty cycle input function.

#### 3 Velocity, acceleration and maximum speed settings

If the source of the speed command is selected as 0-digit given, you need to set the speed given value to be written to object 2006<sub>h</sub> : 04<sub>h</sub> (014B<sub>h</sub>);

If the source of the select speed command is 3-duty cycle given, it is necessary to input pulse signals from DI1 (duty cycle) and DI2 (direction); when the direction input is invalid, 0% duty cycle corresponds to a negative maximum speed value 2000<sub>h</sub> : 0F<sub>h</sub> (006B<sub>h</sub>), and 100% duty cycle corresponds to speed 0; when the input direction is valid, 0% duty cycle corresponds to a positive maximum speed value 2000<sub>h</sub> : 0F<sub>h</sub> (006B<sub>h</sub>), 100% duty cycle corresponds to speed 0;

If the source of main speed command is selected as 4~50% duty cycle speed regulation given, it is necessary to input pulse signal from DI1 (duty cycle), 0~50% corresponds to negative limiting speed 2006<sub>h</sub> : 0B<sub>h</sub> (0152<sub>h</sub>)~0,50%-100% corresponds to 0~positive limiting speed 2006<sub>h</sub> : 0A<sub>h</sub> (0151)<sub>h</sub>

Configuring the speed of acceleration and deceleration 2006<sub>h</sub> : 07<sub>h</sub> (014E<sub>h</sub>) and 2006<sub>h</sub> : 08<sub>h</sub> (014F)<sub>h</sub>

Configuration of the maximum speed 2000<sub>h</sub> : 0F<sub>h</sub> (006B<sub>h</sub>), with a duty cycle regulation of 100% corresponding to the maximum speed

#### 4 (of a computer) run

The motor is enabled according to the set motor enable terminal, and the motor runs at the set speed.

### 5.14.4 Functional Description

In this mode, the motor will run to the appropriate speed based on the input from the terminals or

digital givens.

#### 5.14.5 Example

##### Realization of digitally given speed regulation:

###### 1 Parameter Settings:

- Set speed mode to 2 (NiMotion speed mode): 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>)=2  
Send: 01 10 00 B1 00 01 02 00 02
- Set the main speed command source to 0 (digit given): 2006<sub>h</sub> : 01<sub>h</sub> (0148<sub>h</sub>)=0;  
Send: 01 10 01 48 00 01 02 00 00
- Set the auxiliary speed command source to 0 (number given): 2006<sub>h</sub> : 02<sub>h</sub> (0149<sub>h</sub>)=0;  
Send: 01 10 01 49 00 01 02 00 00
- Set the acceleration ramp time constant to 100ms: 2006<sub>h</sub> : 07<sub>h</sub> (014E<sub>h</sub>)=100  
Send: 01 10 01 4E 00 01 02 00 64
- Set the deceleration ramp time constant to 100ms: 2006<sub>h</sub> : 08<sub>h</sub> (014F<sub>h</sub>)=100  
Send: 01 10 01 4F 00 01 02 00 64
- Set the number given as 60rpm, 2006<sub>h</sub> : 04<sub>h</sub> (014B<sub>h</sub>)=60  
Send: 01 10 01 4B 00 01 02 00 3C
- Terminal Setting  
Set the DI1 terminal:  
Function selection for step enable: 2003<sub>h</sub> : 03<sub>h</sub> (00D5<sub>h</sub>)=1  
Send: 01 10 00 D5 00 01 02 00 01  
Logic select low active: 2003<sub>h</sub> : 04<sub>h</sub> (00D6<sub>h</sub>)=0  
Send: 01 10 00 D6 00 01 02 00 00

###### 2 Running:

- DI1 input is low, the motor is speed-controlled according to the digitally given 2006<sub>h</sub> : 02<sub>h</sub> (0149<sub>h</sub>)

##### Duty cycle input speed regulation is realized:

###### 1 Parameter Settings:

- Set speed mode to 2 (NiMotion speed): 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>)=2  
Send: 01 10 00 B1 00 01 02 00 02
- Set the speed command source to 3 (duty cycle): 2006<sub>h</sub> : 01<sub>h</sub> (0148<sub>h</sub>)=3  
Send: 01 10 01 48 00 01 02 00 03
- Set the acceleration ramp time constant to 100ms: 2006<sub>h</sub> : 07<sub>h</sub> (014E<sub>h</sub>)=100  
Send: 01 10 01 4E 00 01 02 00 64
- Set the deceleration ramp time constant to 100ms: 2006<sub>h</sub> : 08<sub>h</sub> (014F<sub>h</sub>)=100  
Send: 01 10 01 4F 00 01 02 00 64

Set the maximum speed to 3000rpm: 2000<sub>h</sub> : 0F<sub>h</sub> (006B<sub>h</sub>)=3000

Send: 01 10 00 6B 00 01 02 0B B8

- Terminal Setting

DI1 duty cycle input: 2003<sub>h</sub> : 03<sub>h</sub> (00D5<sub>h</sub>)=44

Send: 01 10 00 D5 00 01 02 00 2C

DI2 duty cycle direction input: 2003<sub>h</sub> : 05<sub>h</sub> (00D7<sub>h</sub>)=45

Send: 01 10 00 D7 00 01 02 00 2D

DI3 function number is set to step enable:

Function selection for step enable: 2003<sub>h</sub> : 07<sub>h</sub> (00D9<sub>h</sub>) =1

Send: 01 10 00 D9 00 01 02 00 01

Logic selection active low: 2003<sub>h</sub> : 08<sub>h</sub> (00DA<sub>h</sub>) =0

Send: 01 10 00 DA 00 01 02 00 00

## 2 Running:

- DI1 inputs a square wave with a certain duty cycle, DI3 inputs a low level to enable the motor, and the motor speeds up according to the input duty cycle.

## 5.15 NiMotion Torque Mode

### 5.15.1 Structure Plan

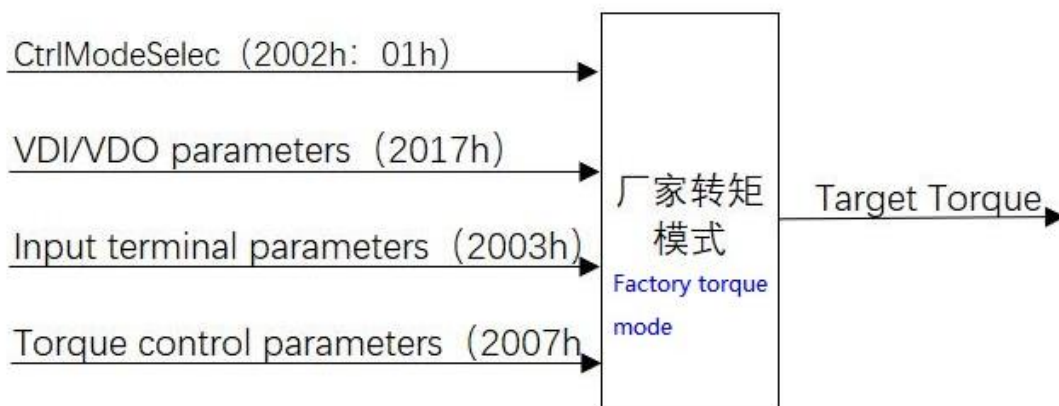


Figure 5- 34

### 5.15.2 Relevant Object

The following objects require attention in this mode:

Table 5- 37

| Object Index                                            | Description                                                                                                    |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 2002 <sub>h</sub> : 01 <sub>h</sub> (00B1) <sub>h</sub> | Operation mode selection                                                                                       |
| 2007 <sub>h</sub> : 01 <sub>h</sub> (015E) <sub>h</sub> | NiMotion torque mode main torque feed source, 0-digital feed                                                   |
| 2007 <sub>h</sub> : 02 <sub>h</sub> (015F) <sub>h</sub> | NiMotion Torque Mode Auxiliary Torque Feed Source, 0-Digital Feed                                              |
| 2007 <sub>h</sub> : 03 <sub>h</sub> (0160) <sub>h</sub> | NiMotion torque mode torque command selection, 0-primary torque given, 1-secondary torque given, 2-primary and |

| Object Index                                            | Description                                                                                                                                                                                                                               |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | secondary torque given (default 0)                                                                                                                                                                                                        |
| 2007 <sub>h</sub> : 04 <sub>h</sub> (0161) <sub>h</sub> | Torque command digital given value (0.1%)                                                                                                                                                                                                 |
| 2007 <sub>h</sub> : 0A <sub>h</sub> (0167) <sub>h</sub> | Positive internal torque limit (0.1%)                                                                                                                                                                                                     |
| 2007 <sub>h</sub> : 0B <sub>h</sub> (0168) <sub>h</sub> | Reverse internal torque limit (0.1%)                                                                                                                                                                                                      |
| 2007 <sub>h</sub> : 10 <sub>h</sub> (016D) <sub>h</sub> | Torque control forward speed limit value (rpm)                                                                                                                                                                                            |
| 2007 <sub>h</sub> : 11 <sub>h</sub> (016E) <sub>h</sub> | Torque control reverse speed limit value (rpm)                                                                                                                                                                                            |
| 2017 <sub>h</sub> (0326h ~0346h)                        | Functional configuration of the virtual input terminals (see: "6.1.1 Digital input")                                                                                                                                                      |
| 2003 <sub>h</sub> (00D5h ~00E6h)                        | Functional configuration of physical input terminals (see "6.1.1 Digital Input" for details)<br>Note: You cannot have multiple input terminals configured as the same switch at the same time, or you will report a terminal setup fault. |

### 5.15.3 Control commands and status messages

There is no state machine management in this mode, the motor is enabled via the terminals.

#### start

Prerequisite for this mode of operation: the integrated low-voltage servo motor is switched to NiMotion torque mode (2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) set to 3).

#### 1 Torque source selection:

Object 2007<sub>h</sub> : 01<sub>h</sub> (015E<sub>h</sub>) is the torque instruction source, meaning: 0-number given;

#### 2 Set the multi-function terminals (both physical and virtual terminals are applicable, see "7.1 Digital Inputs and Outputs" for details on terminal settings):

Configure one input terminal as motor enable, and configure its effective logic level according to the actual situation, the default low level is effective.

#### 3 Torque setting and torque and speed limit settings

Selecting the source of the speed command as digitally given, the torque given value needs to be set and written to object 2007<sub>h</sub> : 04<sub>h</sub> (0161<sub>h</sub>);

Configure positive and negative internal torque limits 2007<sub>h</sub> : 0A<sub>h</sub> (0167<sub>h</sub>) and 2007<sub>h</sub> : 0B<sub>h</sub> (0168)<sub>h</sub>

Configuring forward and reverse speed limit values 2007<sub>h</sub> : 10<sub>h</sub> (016D<sub>h</sub>) and 2007<sub>h</sub> : 11<sub>h</sub> (016E)<sub>h</sub>

#### 4 Run

The motor is enabled according to the set motor enable terminal, and the motor operates at the set torque.

#### 5.15.4 Functional Description

In this mode, the motor will run to the corresponding torque by means of the enable control and torque digitalization of the physical or virtual terminals. When the motor's operating speed exceeds the value set by the speed limits (forward speed limit value  $2007_h : 10_h$  (016D<sub>h</sub>) and reverse speed limit value  $2007_h : 11_h$  (016E<sub>h</sub>)), it automatically switches to speed mode control and controls the current speed within the speed limits. When the given target torque is detected to be less than the average torque at the current speed, the motor exits speed control and reverts to torque mode control.

#### 5.15.5 Example

##### 1. Parameter Settings:

- Set the speed mode to 3 (NiMotion torque mode):  $2002_h : 01_h$  (00B1<sub>h</sub>)=3;  
Send: 01 10 00 B1 00 01 02 00 03
- Set the speed command source to 0 (number given):  $2007_h : 01_h$  (015E<sub>h</sub>)=0  
Send: 01 10 01 5E 00 01 02 00 00
- Setting the speed command selection as the main torque command:  $2007_h : 03_h$  (0160<sub>h</sub>)=0  
Send: 01 10 01 60 00 01 02 00 00
- Set the positive and negative internal torque limits to 100%:  $2007_h : 0A_h$  (0167<sub>h</sub>)=1000 and  $2007_h : 0B_h$  (0168<sub>h</sub>)=1000  
Send: 01 10 01 67 00 01 02 03 E8 (sets positive internal torque limit to 100%)  
Send: 01 10 01 68 00 01 02 03 E8 (sets reverse internal torque limit to 100%)  
Set the forward and reverse speed limit values to 3000rpm:  $2007_h : 10_h$  (016D<sub>h</sub>)=3000 and  $2007_h : 11_h$  (016E<sub>h</sub>)=3000  
Send: 01 10 01 6D 00 01 02 0B B8 (set forward speed limit value to 3000rpm)  
Send: 01 10 01 6E 00 01 02 0B B8 (set reverse speed limit value to 3000rpm)
- Setting the numeric value to 100%,  $2007_h : 04_h$  (0161<sub>h</sub>)=1000  
Send: 01 10 01 61 00 01 02 03 E8
- Terminal Setting  
Set the DI1 terminal:  
Function selection for step enable:  $2003_h : 03_h$  (00D5<sub>h</sub>)=1  
Send: 01 10 00 D5 00 01 02 00 01  
  
Logic selection active low:  $2003_h : 04_h$  (00D6<sub>h</sub>)=0  
Send: 01 10 00 D6 00 01 02 00 00

##### 2. Running:

- DI1 input is low and the motor runs at the torque digitally given  $2007_h : 04_h$  (0161<sub>h</sub>)



## 5.16 Multi-Segment Position Mode

### 5.16.1 Structure Plan

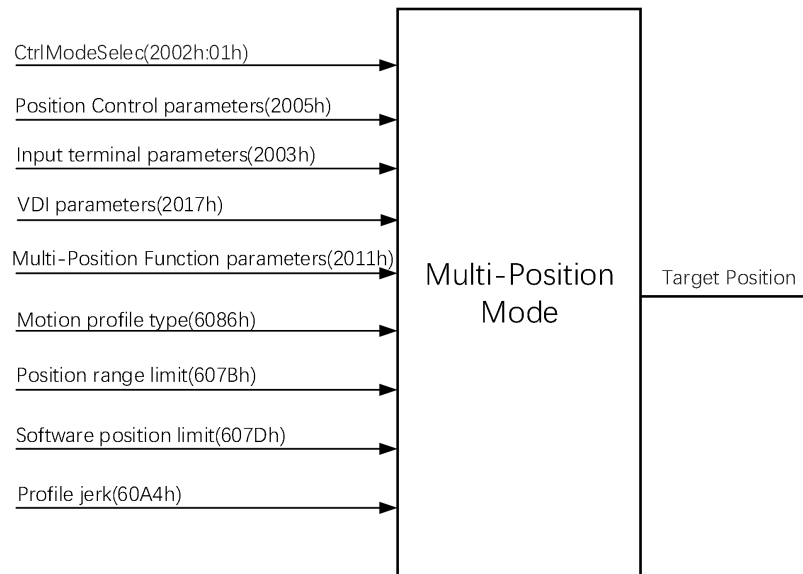


Figure 5- 35

### 5.16.2 Relevant Object

The following objects require attention in this mode:

Table 5- 38

| Object Index                                              | Description                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2002 <sub>h</sub> : 01 <sub>h</sub> (00B1 ) <sub>h</sub>  | Operation mode selection                                                                                                                                                                                                                                                                                                                                                                              |
| 2005 <sub>h</sub> : 01 <sub>h</sub> (010D ) <sub>h</sub>  | Position command source, 0-pulse, 1-step, 2-multi-segment position                                                                                                                                                                                                                                                                                                                                    |
| 2005 <sub>h</sub> : 05 <sub>h</sub> (0112 ) <sub>h</sub>  | Step amount setting (encoder unit)                                                                                                                                                                                                                                                                                                                                                                    |
| 2017 <sub>h</sub> (0326 <sub>h</sub> ~0346 ) <sub>h</sub> | Functional configuration of the virtual input terminals (see: "6.1.1 Digital input")                                                                                                                                                                                                                                                                                                                  |
| 2003 <sub>h</sub> (00D5 <sub>h</sub> ~00E6 ) <sub>h</sub> | Functional configuration of the physical input terminals (see: "6.1.1 Digital input")<br>Note: You cannot have multiple input terminals configured as the same switch at the same time, or you will report a terminal setup fault.                                                                                                                                                                    |
| 2011 <sub>h</sub> (0291 <sub>h</sub> ~02E6 ) <sub>h</sub> | Multi-stage position function parameter set                                                                                                                                                                                                                                                                                                                                                           |
| 6086 <sub>h</sub> (0402 ) <sub>h</sub>                    | Selection of the type of planner curve, i.e. slope.<br>If the value is "0", then there will be no limitation of shocks (plus acceleration), i.e. a trapezoidal curve;<br>If the value is "3", the shock (plus acceleration) will be limited by the value of 60A4 <sub>h</sub> : 01 <sub>h</sub> -02 <sub>h</sub> (041E <sub>h</sub> ~0420 <sub>h</sub> ), i.e., S-curve, and the device only uses the |

|                                                           |                                                                                         |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------|
|                                                           | indexes 01 <sub>h</sub> and 02 <sub>h</sub> (041E <sub>h</sub> and 0420 <sub>h</sub> ). |
| 607D <sub>h</sub> (03EF <sub>h</sub> ~03F1 <sub>h</sub> ) | Limit values for target position (user units)                                           |
| 60A4 <sub>h</sub> (041E <sub>h</sub> ~0424 <sub>h</sub> ) | Contour plus acceleration, see 6.3.2 for details of the implementation process.         |

### 5.16.3 Control Commands and Status Messages

There is no state machine management in this mode, the motor is enabled via the terminals.

#### Enable

Prerequisite for operation in this mode: Motor operation mode is switched to NiMotion position mode (2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>) write 1).

#### 1. Location Source Selection:

Object 2005<sub>h</sub> :01<sub>h</sub> (010D<sub>h</sub>) is the source of position instructions, 2005<sub>h</sub> :01<sub>h</sub> (010D<sub>h</sub>) Meaning: 0-pulse, 1-step amount, 2-multiple positions;

#### 2. Setting the multi-function terminals (both physical and virtual terminals are applicable, see "6.1 Digital Inputs and Outputs" for details on terminal settings).

Set the position command source as 2, configure one input terminal as motor enable, and configure its valid logic level according to the actual situation, with the default low level valid;

Set the other input terminal as multi-segment position/speed enable (Function No. 28), and configure its valid logic level according to the actual situation, the default is valid low;

#### 3. Position Setting

The corresponding operating mode and operating parameters for each segment are set via the multi-segment position function parameter set 2011<sub>h</sub> (0291<sub>h</sub> ~02E6<sub>h</sub>).

#### 4. (of a computer) run

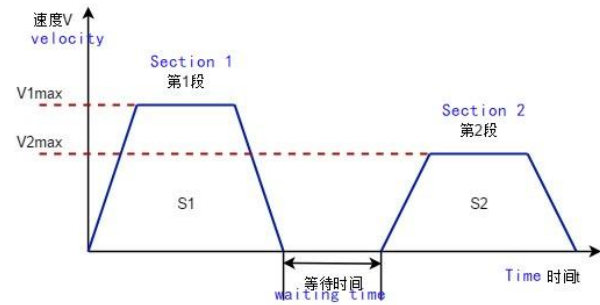
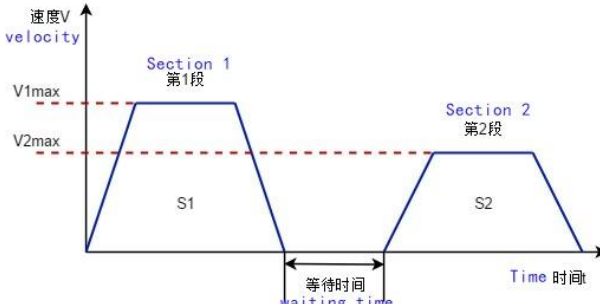
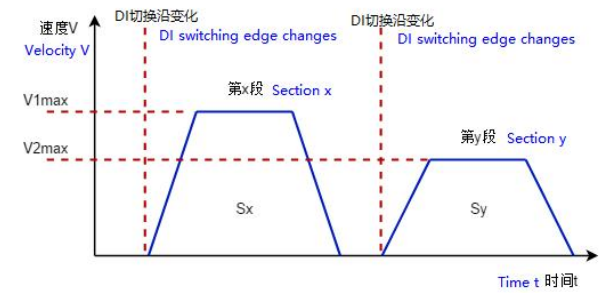
The motor is first enabled via one terminal, and then the multi-segment position instruction is enabled via the other terminal, and the motor runs in accordance with the operation mode and the operation parameters of each segment set in the multi-segment position function parameter set 2011<sub>h</sub> (0291<sub>h</sub> ~02E6<sub>h</sub>).

### 5.16.4 Functional Description

The multi-segment position mode has 3 types of operation, namely, single operation, cyclic operation and DI switching operation. When using the multi-segment position function, the DI port must be set to enable (function No. 1) and multi-segment position/speed enable (function No. 28) respectively; during each segment operation, it must be ensured that the enable is valid; in the non-DI switching operation mode, in addition to the enable being valid, it must be ensured that the multi-segment position/speed enable is valid, and if the multi-segment position/speed enable is turned off, it will give up the displacement commands that have not been sent in this segment and stop the machine.

Specific descriptions of each mode of operation are shown in the table below:

Table 5- 39

| Operation mode                                                                                                                                                                                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Running curve                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| single run<br>2011 <sub>h</sub> :01 <sub>h</sub><br>(0291 <sub>h</sub> )=0                                                                                                                                                                         | Paragraphs 1 through 16 are run once;<br>Automatic incremental switching of segment numbers;<br>Waiting time can be set between segments;<br>Multi-segment position/speed enable is level active.                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| run in a loop<br>2011 <sub>h</sub> :01 <sub>h</sub><br>(0291 <sub>h</sub> )=1                                                                                                                                                                      | 2011 <sub>h</sub> :06 <sub>h</sub> (0296 <sub>h</sub> ) The number of times the starting segment set by 2011 :06 (0296) has run the end segment set by 2011 <sub>h</sub> :02 <sub>h</sub> (0292 <sub>h</sub> ) the number of times the 2011 <sub>h</sub> :04 <sub>h</sub> (0294 <sub>h</sub> ) has run the 2011 :04 (0294) set by 2011 :02 (0292);<br>Automatic incremental switching of segment numbers;<br>Waiting time can be set between segments;<br>Multi-segment position/speed enable is level active;<br>Note: If the setting value of the end segment is smaller than the start segment, segments 1 through 16 cycle once. |   |
| DI switching operation<br>2011 <sub>h</sub> :01 <sub>h</sub><br>(0291 <sub>h</sub> )=2<br><br><b>Note: Physical and virtual terminals can be used in combination and the combination must be 4-digit; for example: Physical terminal selection</b> | Position segment number switching is determined by the decimal 0 to 15 corresponding to the 4-digit binary number consisting of multifunction numbers 6, 7, 8 and 9;<br>The wait time set between segments is invalid;<br>Multi-segment position/speed enable is level active;<br>The relationship between segment number and multifunction number corresponds as follows:                                                                                                                                                                                                                                                           |  |

| Operation mode                                    | Description |          |          |          |                           | Running curve |
|---------------------------------------------------|-------------|----------|----------|----------|---------------------------|---------------|
| DI1=6,DI2=7,<br>Virtual terminals<br>VI1=8,VI2=9. | Fu<br>n9    | Fu<br>n8 | Fu<br>n7 | Fu<br>n6 | secti<br>on<br>num<br>ber |               |
|                                                   | 0           | 0        | 0        | 0        | 1                         |               |
|                                                   | 0           | 0        | 0        | 1        | 2                         |               |
|                                                   | .....       |          |          |          |                           |               |
|                                                   | 1           | 1        | 1        | 1        | 16                        |               |

### 5.16.5 Configuration Example

#### Multi-stage position-single run control

- Parameter setting

- Set to NiMotion position mode: 2002<sub>h</sub> : 01<sub>h</sub> (00B1<sub>h</sub>)=1

Send: 01 10 00 B1 00 01 02 00 01

Set the source of the position command to a multi-segment position: 2005<sub>h</sub> :01<sub>h</sub> (010D<sub>h</sub>)=2;

Send: 01 10 01 0D 00 01 02 00 02

Sets the multi-segment position function parameter group:

Segment position run mode: 2011<sub>h</sub> :01<sub>h</sub> (0291<sub>h</sub>) = 0 (single run);

Send: 01 10 02 91 00 01 02 00 00

Position instruction type selection: 2011<sub>h</sub> :05<sub>h</sub> (0295<sub>h</sub>) = 0 (relative displacement);

Send: 01 10 02 95 00 01 02 00 00

Operational parameters for segment 1 displacements:

Moving Displacement 2011<sub>h</sub> :07<sub>h</sub> (0297<sub>h</sub>) = 10000 (user units), Maximum Allowable Speed 2011<sub>h</sub> :08<sub>h</sub> (0299<sub>h</sub>) = 500 (in rpm).

Send: 01 10 02 97 00 02 04 00 00 27 10 (set first segment displacement to 10000)

Send: 01 10 02 99 00 01 02 01 F4 (set the maximum operating speed of the first displacement to 500 rpm)

Acceleration and deceleration time 2011<sub>h</sub> :09<sub>h</sub> (029A<sub>h</sub>) = 500 (in ms), waiting time after displacement completion 2011<sub>h</sub> :0A<sub>h</sub> (029B<sub>h</sub>) = 1000 (in ms)

Send: 01 10 02 9A 00 01 02 01 F4 (set acceleration/deceleration time to 500ms)

Send: 01 10 02 9B 00 01 02 03 E8 (set the wait time after the first displacement is completed to 1000ms)

Paragraphs 2 to 16 are set with reference to the operational parameters of paragraph 1 displacement;

Set motor enable: 2003<sub>h</sub> :03<sub>h</sub> (00D5<sub>h</sub>)=1, logic active low: 2003<sub>h</sub> :04<sub>h</sub> (00D6<sub>h</sub>)=0;

Send: 01 10 00 D5 00 01 02 00 01 (set DI1 to enable)